

Lecture 12 – Wind Energy

Part II: Wind Power Statistics

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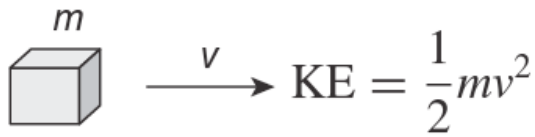
ECE, UCSC



Outline

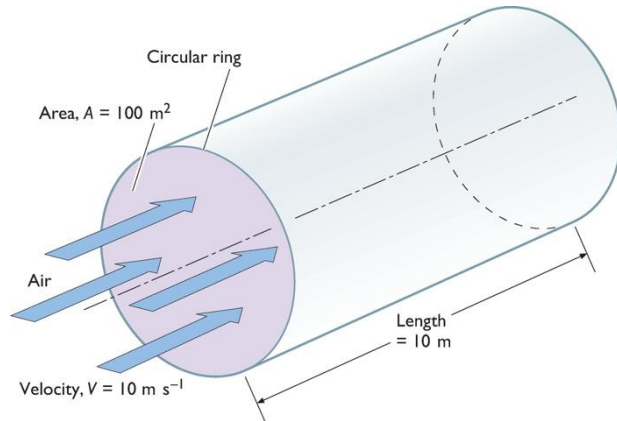
- Air Density: Impact of Temperature and Height
 - The Betz Limit
 - Speed-to-Power Mapping
 - Wind Histogram and PDF
 - Weibull and Rayleigh Distributions
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- B3 Chap 8
 - B1 Chap 7

Energy/Power in Wind



moving air

kinetic energy	=	kilograms metres per second (ms⁻¹) half mass x velocity squared	=	0.5mV ²
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Mass of moving air per second (mass flow rate)
 = air density x area x length of cylinder of air passing per second
 = air density x area x velocity

$$\text{mass flow rate (mass/time)} = \rho AV$$

$$\Rightarrow P = 0.5\rho AV^3 \quad \text{Watt} = \frac{\text{kg}}{\text{m}^3} \times \text{m}^2 \times \left(\frac{\text{m}}{\text{s}}\right)^3$$

- Air density → linear effect
- Blade diameter → quadratic effect
- Wind speed → cubic effect

Cube Law

$$\frac{P}{A} = 0.5\rho V^3$$

Windspeed (m/s)	Windspeed (mph)	Power (W/m ²)
0	0	0
1	2.24	1
2	4.47	5
3	6.71	17
4	8.95	39
5	11.19	77
6	13.42	132
7	15.66	210
8	17.90	314
9	20.13	447
10	22.37	613
11	24.61	815
12	26.84	1058
13	29.08	1346
14	31.32	1681
15	33.56	2067

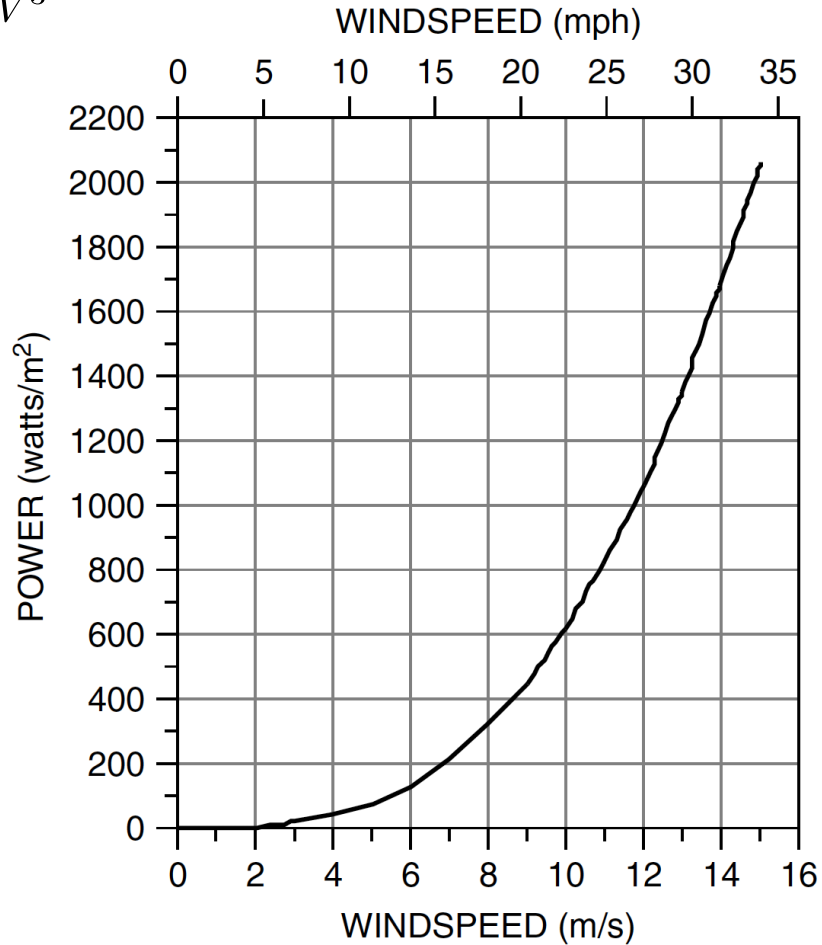


Fig: Power in the wind per square meter of cross section (15 °C and 1 atm).

ρ is the air density (kg/m³); at 15 °C and 1 atm, $\rho = 1.225$ kg/m³.

1 atm = 1.01325 bar = 101.325 kPa

Example

Example 7.1 Don't Just Use Average Wind Speed. Compare the amount of wind energy at 15°C and 1-atm pressure that passes through 1 m² of cross-sectional area for the following wind regimes:

- a. 100 h of 6 m/s winds (13.4 mph)
- b. 50 h at 3 m/s plus 50 h at 9 m/s (i.e., an average wind speed of 6 m/s)

Solution

- a. With steady 6 m/s winds, all we have to do is multiply power, given in Equation 7.7, by hours:

$$\begin{aligned}\text{Energy (6 m/s)} &= \frac{1}{2} \rho A v^3 \Delta t \\ &= \frac{1}{2} \cdot 1.225 \text{ kg/m}^3 \cdot 1 \text{ m}^2 \cdot (6 \text{ m/s})^3 \cdot 100 \text{ h} = 13,230 \text{ Wh}\end{aligned}$$

- b. With 50 h at 3 m/s

$$\text{Energy (3 m/s)} = \frac{1}{2} \cdot 1.225 \text{ kg/m}^3 \cdot 1 \text{ m}^2 \cdot (3 \text{ m/s})^3 \cdot 50 \text{ h} = 827 \text{ Wh}$$

And 50 h at 9 m/s contains

$$\text{Energy (9 m/s)} = \frac{1}{2} \cdot 1.225 \text{ kg/m}^3 \cdot 1 \text{ m}^2 \cdot (9 \text{ m/s})^3 \cdot 50 \text{ h} = 22,326 \text{ Wh}$$

for a total of $827 + 22,326 = 23,153 \text{ Wh}$

Temperature Correction for Air Density

The ideal gas law: $PV = nRT$ n is the amount of substance of gas (aka. number of moles).

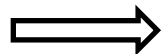
where P is the absolute pressure (atm), V is the volume (m^3), n is the mass (mol), R is the ideal gas constant $= 8.2056 \times 10^{-5} \text{ m}^3 \cdot \text{atm} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$, and T is the absolute temperature (K), where $K = ^\circ\text{C} + 273.15$.

If we let M.W. stand for the molecular weight of the gas (g/mol), we can write the following expression for air density, ρ :

$$\rho \text{ (kg/m}^3\text{)} = \frac{n(\text{mol}) \cdot \text{M.W. (g/mol)} \cdot 10^{-3} \text{ (kg/g)}}{V \text{ (m}^3\text{)}}$$

$$n = \frac{PV}{RT}$$

$$\Rightarrow \rho = \frac{P \times \text{M.W.} \times 10^{-3}}{RT} = \frac{p(\text{atm})}{T(\text{K})} \times \frac{28.97 \text{ g/mol} \times 10^{-3} \text{ kg/g}}{8.2056 \times 10^{-5} \text{ m}^3 \cdot \text{atm}/(\text{K} \cdot \text{mol})}$$



$$\rho(\text{kg/m}^3) = \frac{353.1 p(\text{atm})}{T(\text{K})}$$

Temperature Correction for Air Density (cont'd)

Since we are working with air, we can easily figure out its equivalent molecular weight by looking at its constituent molecules, which are mostly nitrogen (78.08%), oxygen (20.95%), a little bit of argon (0.93%), carbon dioxide (0.039%), and so forth. Using the constituent molecular weights ($N_2 = 28.02$, $O_2 = 32.00$, $Ar = 39.95$, $CO_2 = 44.01$) we find the equivalent molecular weight of air to be

$$\begin{aligned} \text{M.W.}(\text{air}) &= 0.781 \times 28.02 + 0.2095 \times 32.00 \\ &\quad + 0.0093 \times 39.95 + 0.00039 \times 44.01 = 28.97 \end{aligned}$$

Q: Find the density of air at 1 atm and 30 °C

Solution.

$$\rho = \frac{1 \text{ atm} \times 28.97 \text{ g/mol} \times 10^{-3} \text{ kg/g}}{8.2056 \times 10^{-5} \text{ m}^3 \cdot \text{atm}/(\text{K} \cdot \text{mol}) \times (273.15 + 30) \text{ K}} = 1.165 \text{ kg/m}^3$$

Altitude Correction for Air Density

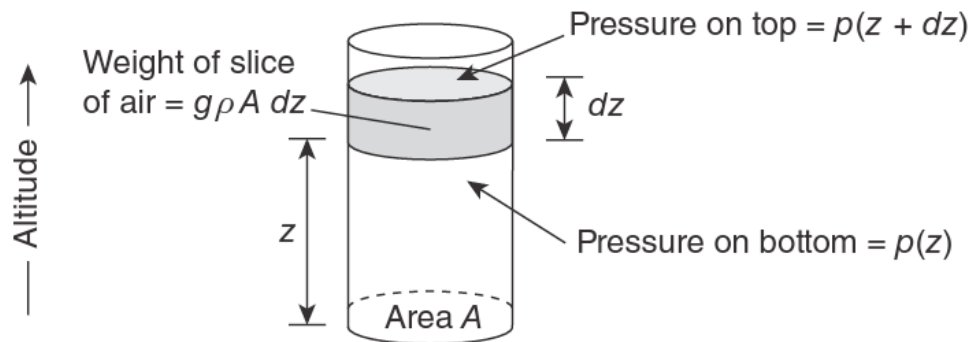


Fig: A column of air in static equilibrium used to determine the relationship between air pressure and altitude.

$$p(z) = p(z + dz) + \frac{g\rho A dz}{A}$$

Express the incremental pressure dp for an incremental change in elevation dz

$$dp = p(z + dz) - p(z) = -g\rho dz$$

$$\text{That is, } \frac{dp}{dz} = -\rho g$$

Altitude Correction for Air Density (cont'd)

$$\begin{aligned}\frac{dp}{dz} &= -\rho g = -\frac{353.1}{T} \left(\frac{\text{kg}}{\text{m}^3} \right) \times \left(9.806 \frac{\text{m}}{\text{s}^2} \right) \left(\frac{\text{atm}}{101,325 \text{ Pa}} \times \frac{1 \text{ Pa}}{\text{N/m}^2} \times \frac{1 \text{ N}}{\text{kg} \cdot \text{m/s}^2} \right) \cdot p \\ &= -\left(\frac{0.0342}{T} \right) p\end{aligned}$$

$$\Rightarrow p = p_0 \exp(-0.0342z/T)$$

where p_0 is the reference pressure of 1 atm.

Finally, we get a useful density correction for both temperature and altitude:

$$\rho(\text{kg/m}^3) = \frac{353.1 \exp(-0.0342 z/T)}{T}$$

where T is in kelvins (K) and z is in meters.

Quiz

$$\rho(\text{kg/m}^3) = \frac{353.1 \exp(-0.0342 z/T)}{T}$$

Example 7.2 Combined Temperature and Altitude Correction. Find the power density (W/m^2) in 10 m/s winds at an elevation of 2000 m (6562 ft) and a temperature of 25°C (298.15 K). Compare that to the power density under standard 1-atm and 15°C conditions.

Solution. From Equation 7.17

$$\rho = \frac{353.1 \exp(-0.0342 \times 2000/298.15)}{298.15} = 0.9415 \text{ kg/m}^3$$

$$P/A = \frac{1}{2} \rho v^3 = 0.5 \times 0.9415 \times 10^3 = 470.8 \text{ W/m}^2$$

Under standard conditions

$$P/A = \frac{1}{2} \rho v^3 = 0.5 \times 1.225 \times 10^3 = 612.5 \text{ W/m}^2$$

That is a 23% decrease in power density at the higher elevation and temperature.

Impact of Tower Height

- In the first few hundred meters above the ground, wind speed is greatly affected by the friction that the air experiences as it moves across the earth's surface.
- Smooth surfaces, such as a calm sea, offer very little resistance, and the variation of speed with elevation is modest.
- At the other extreme, surface winds are slowed considerably by high irregularities such as forests and buildings.

$$\left(\frac{v}{v_0} \right) = \left(\frac{H}{H_0} \right)^\alpha$$

where v is the windspeed at height H , v_0 is the windspeed at height H_0 (often a reference height of 10 m), and α is the friction coefficient.

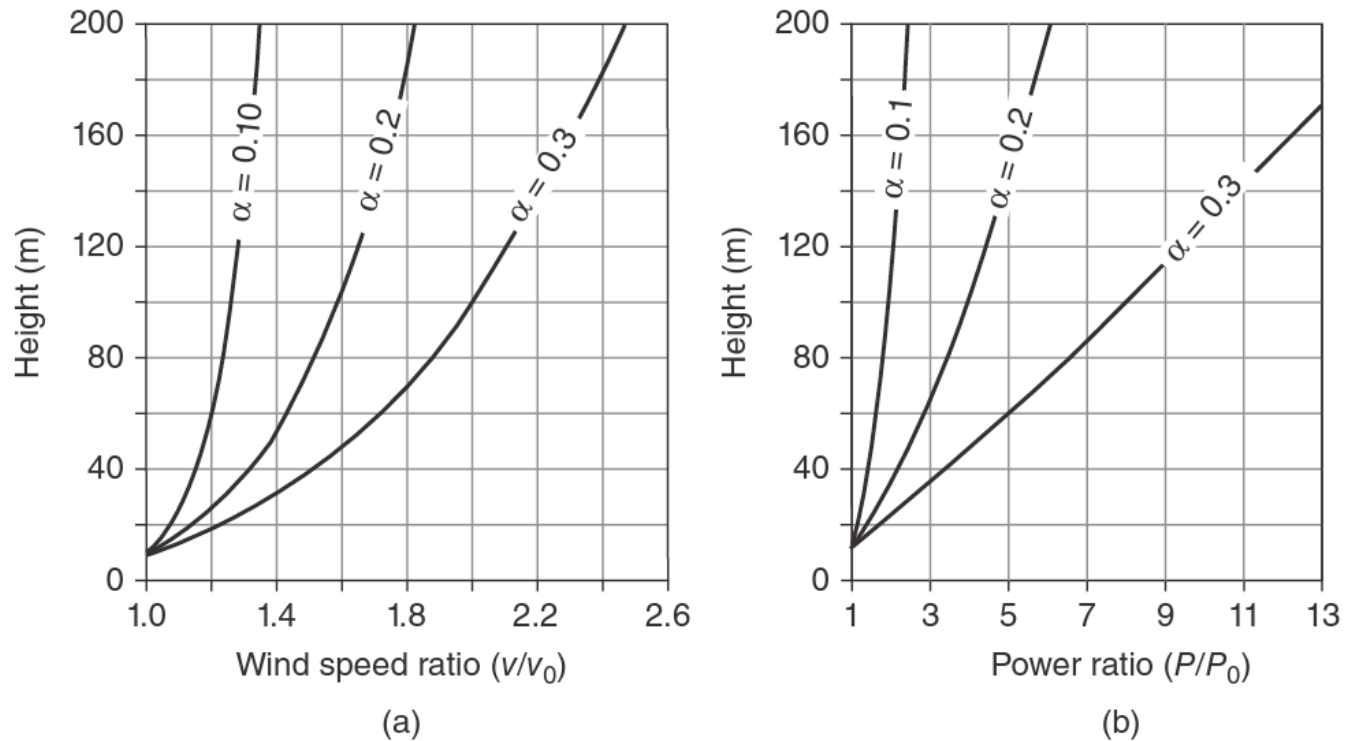
Impact of Tower Height (cont'd)

α is a function of the terrain over which the wind blows.

Terrain Characteristics	Friction Coefficient α
Smooth hard ground, calm water	0.10
Tall grass on level ground	0.15
High crops, hedges and shrubs	0.20
Wooded countryside, many trees	0.25
Small town with trees and shrubs	0.30
Large city with tall buildings	0.40

Oftentimes, for rough approximations in somewhat open terrain a value of $1/7$ (the “one-seventh” rule-of-thumb) is used for α .

Impact of Tower Height (cont'd)



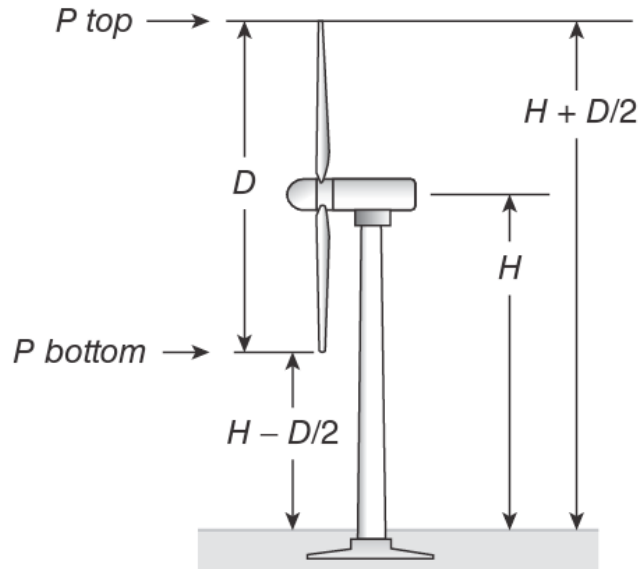
Increasing (a) windspeed and (b) power ratios with height for various friction coefficients α using a reference height of 10 m

$$\left(\frac{P}{P_0}\right) = \left(\frac{1/2\rho A v^3}{1/2\rho A v_0^3}\right) = \left(\frac{v}{v_0}\right)^3 = \left(\frac{H}{H_0}\right)^{3\alpha}$$

Quiz

Example 7.3 Impact of Tower Height on Rotor Stress. A wind turbine with a 50-m rotor diameter is to be mounted on either a 50-m tower or an 80-m tower. Assume the usual 1/7th rule of thumb for the shear-friction coefficient.

- Compare the wind power density at each hub height.
- For each height, compare the ratio of the power density at the highest point that the tip of a rotor blade reaches to the lowest point to which it falls.



- Using Equation 7.20 with 50-m and 100-m hub heights:

$$\frac{P}{P_0} = \left(\frac{H}{H_0} \right)^{3\alpha} = \left(\frac{80}{50} \right)^{3 \times 1/7} = 1.22$$

so there is 22% more power available at the taller hub height.

- For the 50-m blade at a 50-m hub height, the tip reaches 75 m at its highest point and 25 m at its lowest:

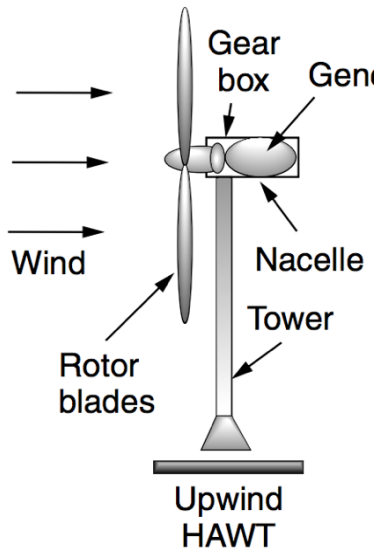
$$\frac{P}{P_0} = \left(\frac{75}{25} \right)^{3 \times 1/7} = 1.60$$

The power in the wind at the top of the rotor swing is 60% higher than it is when the tip reaches its lowest point.

At the 80-m hub height, the power ratio from the highest to the lowest point will be

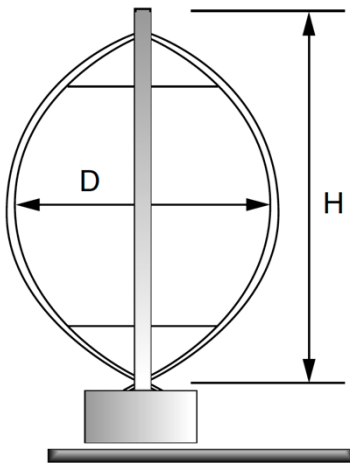
$$\frac{P}{P_0} = \left(\frac{105}{55} \right)^{3 \times 1/7} = 1.32$$

Effect of Swept Area



$$A = (\pi/4) D^2$$

- Doubling the diameter increases the power available by a factor of four.
- Turbine cost increases roughly in proportion to blade diameter, but power is proportional to diameter squared.
- Bigger machines have proven to be more cost effective.

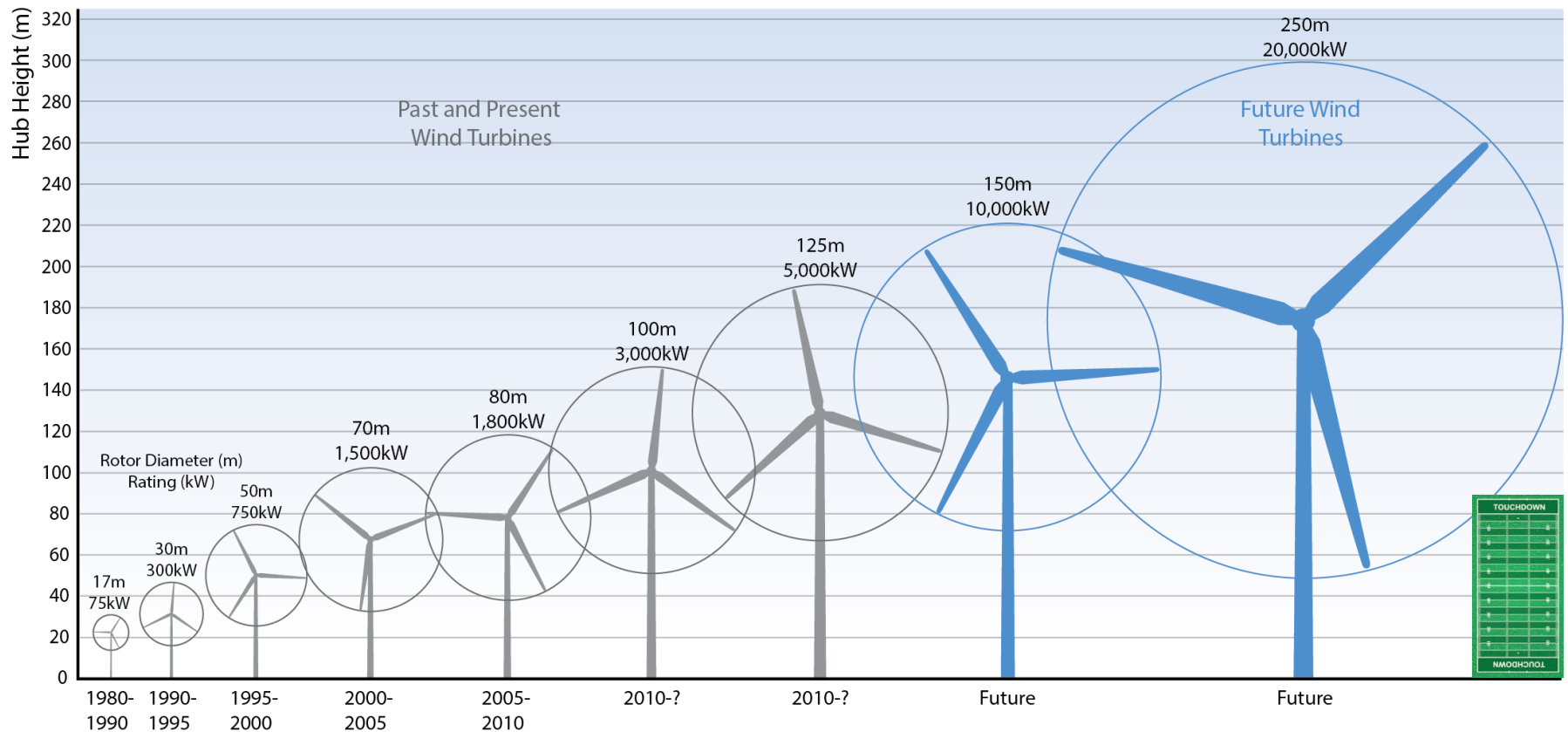


$$A \cong \frac{2}{3} D \cdot H$$

Vertical-axis Darrieus rotor

Development of Wind Power (Cont'd)

Wind Farm Growth Through the Years



Wind Power Conversion Efficiency

Power in the wind passing through a wind turbine rotor is proportional to:

- density of the air
- area of the rotor
- cube of the wind velocity

The power contained in the wind is not in practice the amount of power that can be extracted by a wind turbine because **losses** are incurred in the energy **extraction/conversion** process.

Maximum Rotor Efficiency

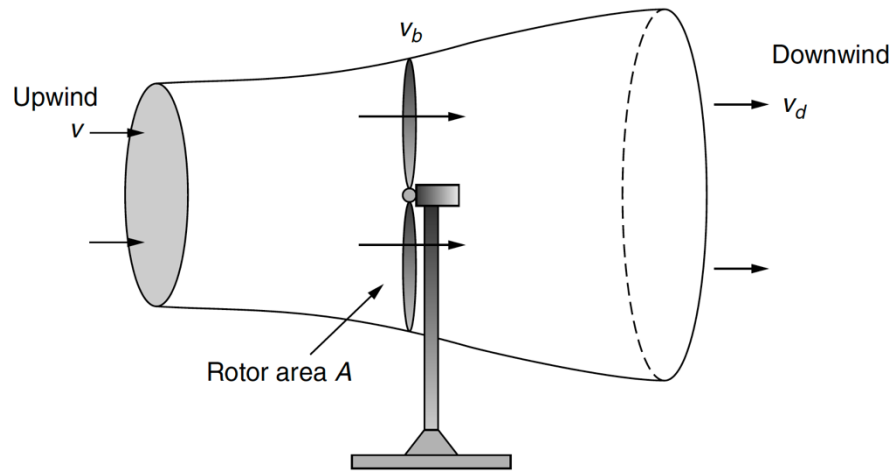


Fig: Approaching wind slows and expands as a portion of its kinetic energy is extracted by the wind turbine. The envelope forms a “*stream tube*”.

The power extracted by the blades P_b is equal to the difference in kinetic energy between the upwind and downwind air flows

$$P_b = \frac{1}{2} \dot{m} (v^2 - v_d^2)$$

Mass flow rate is the mass of a substance which passes per unit of time (kg/s):

$$\dot{m} = \lim_{\Delta t \rightarrow 0} \frac{\Delta m}{\Delta t} = \frac{dm}{dt}$$

The mass flow rate of air within the stream tube is everywhere the same.

$$\dot{m} = \rho A v_b$$

Betz Efficiency

$$\text{Assume } v_b = \frac{1}{2}(v + v_d) \Rightarrow P_b = \frac{1}{2}\rho A \left(\frac{v + v_d}{2} \right) (v^2 - v_d^2)$$

The ratio of downstream-to-upstream windspeed $\lambda = \left(\frac{v_d}{v} \right)$

$$P_b = \frac{1}{2}\rho A \left(\frac{v + \lambda v}{2} \right) (v^2 - \lambda^2 v^2) = \underbrace{\frac{1}{2}\rho A v^3}_{\text{Power in the wind}} \cdot \underbrace{\left[\frac{1}{2}(1 + \lambda)(1 - \lambda^2) \right]}_{\text{Fraction extracted}}$$

$$\text{Rotor efficiency} = C_P = \frac{1}{2}(1 + \lambda)(1 - \lambda^2)$$

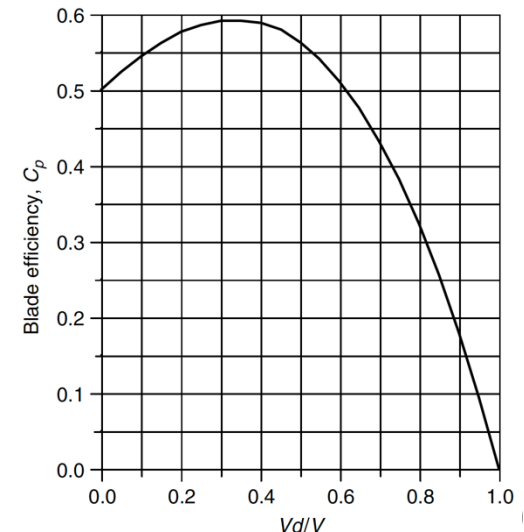
$$\frac{dC_P}{d\lambda} = \frac{1}{2}[(1 + \lambda)(-2\lambda) + (1 - \lambda^2)] = 0$$

which has solution

$$\lambda = \frac{v_d}{v} = \frac{1}{3}$$

$$\text{Maximum rotor efficiency} = \frac{1}{2} \left(1 + \frac{1}{3} \right) \left(1 - \frac{1}{3^2} \right) = 59.3\%$$

Established By German physicist Albert Betz'1919



Tip-Speed-Ratio

- If the rotor turns too slowly, the efficiency drops off since the blades are letting too much wind pass by unaffected.
- If the rotor turns too fast, efficiency is reduced as the turbulence caused by one blade increasingly affects the blade that follows.

tangential speed

$$\text{Tip-Speed-Ratio (TSR)} = \frac{\text{Rotor tip speed}}{\text{Wind speed}} = \frac{\text{rpm} \times \pi D}{60 v}$$

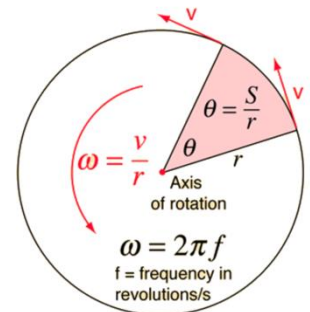
where rpm is the rotor speed, revolutions per minute; D is the rotor diameter (m); and v is the wind speed (m/s) upwind of the turbine.

- Revolutions per minute (rpm) is the number of turns in one minute.

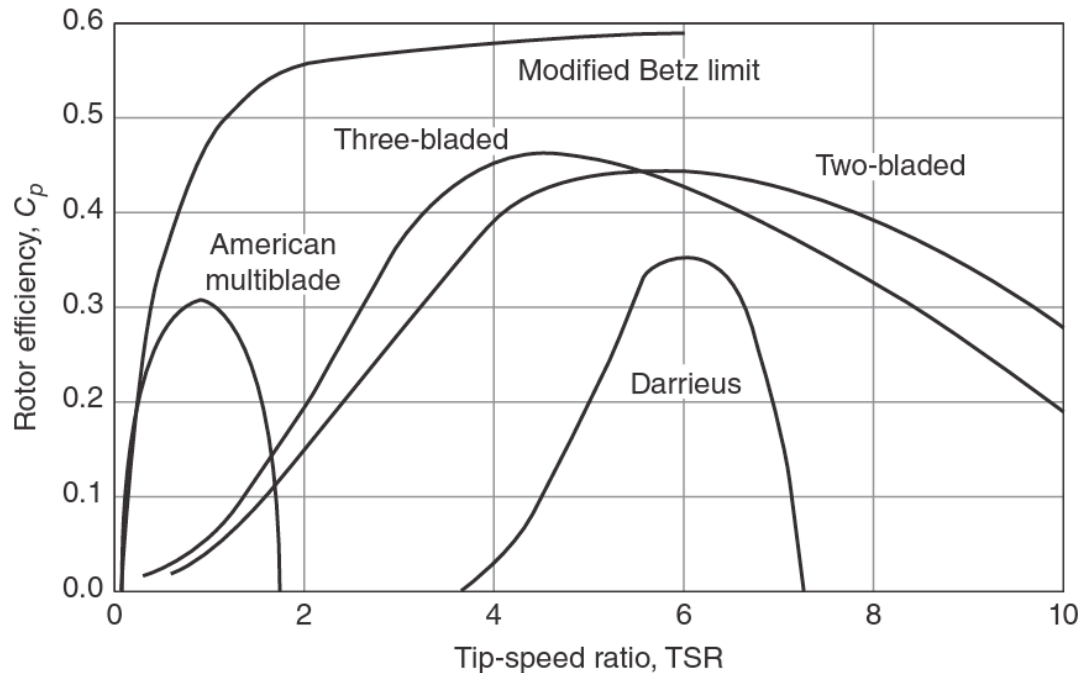
$$1 \text{ rad/s} = \frac{1}{2\pi} \text{ Hz} = \frac{60}{2\pi} \text{ rpm}$$

angular speed

rpm: rotational speed



Rotor Types vs TSR



Rotors with fewer blades reach their optimum efficiency at higher rotational speeds.

- American multiblade spins slowly, maximum efficiency about 30%.
- The two- and three-blade rotors spin much faster, maximum efficiencies of roughly 40–50%.
- The “ideal efficiency” line reflects the fact that a slowly turning rotor does not intercept all of the wind, which reduces the maximum possible efficiency to something below the Betz limit.

Example (1/2)

Example 6.7 How Fast Does a Big Wind Turbine Turn? A 40-m, three-bladed wind turbine produces 600 kW at a windspeed of 14 m/s. Air density is the standard 1.225 kg/m³. Under these conditions,

- At what rpm does the rotor turn when it operates with a TSR of 4.0?
- What is the tip speed of the rotor?
- If the generator needs to turn at 1800 rpm, what gear ratio is needed to match the rotor speed to the generator speed?
- What is the efficiency of the complete wind turbine (blades, gear box, generator) under these conditions?

Solution

- Using (6.27),

$$\text{rpm} = \frac{\text{TSR} \times 60 \, v}{\pi D} = \frac{4 \times 60 \, \text{s/min} \times 14 \, \text{m/s}}{40\pi \, \text{m/rev}} = 26.7 \, \text{rev/min}$$

That's about 2.2 seconds per revolution ... pretty slow!

Example (2/2)

b. The tip of each blade is moving at

$$\text{Tip speed} = \frac{26.7 \text{ rev/min} \times \pi 40 \text{ m/rev}}{60 \text{ s/min}} = 55.9 \text{ m/s}$$

Notice that even though 2.2 s/rev sounds slow; the tip of the blade is moving at a rapid 55.9 m/s, or 125 mph.

c. If the generator needs to spin at 1800 rpm, then the gear box in the nacelle must increase the rotor shaft speed by a factor of

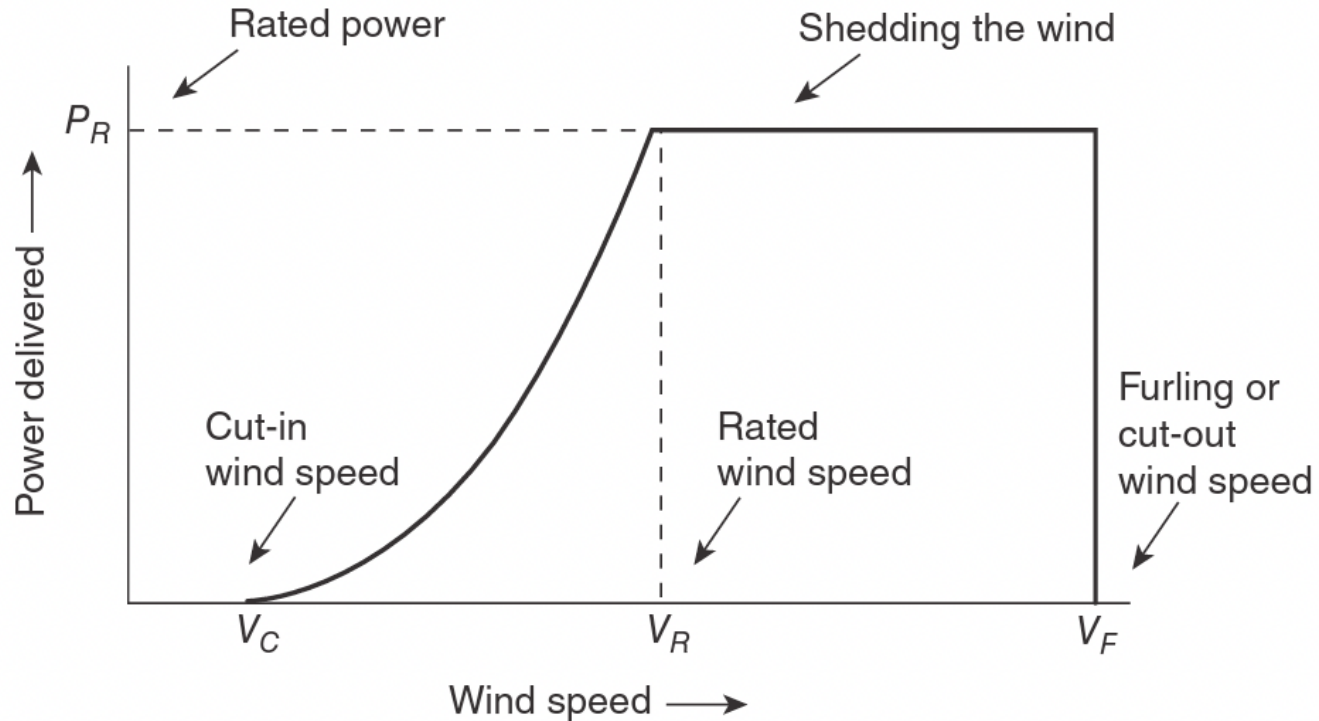
$$\text{Gear ratio} = \frac{\text{Generator rpm}}{\text{Rotor rpm}} = \frac{1800}{26.7} = 67.4$$

d. From (6.4) the power in the wind is

$$P_w = \frac{1}{2} \rho A v_w^3 = \frac{1}{2} \times 1.225 \times \frac{\pi}{4} \times 40^2 \times 14^3 = 2112 \text{ kW}$$

$$\text{Overall efficiency} = \frac{600 \text{ kW}}{2112 \text{ kW}} = 0.284 = 28.4\%$$

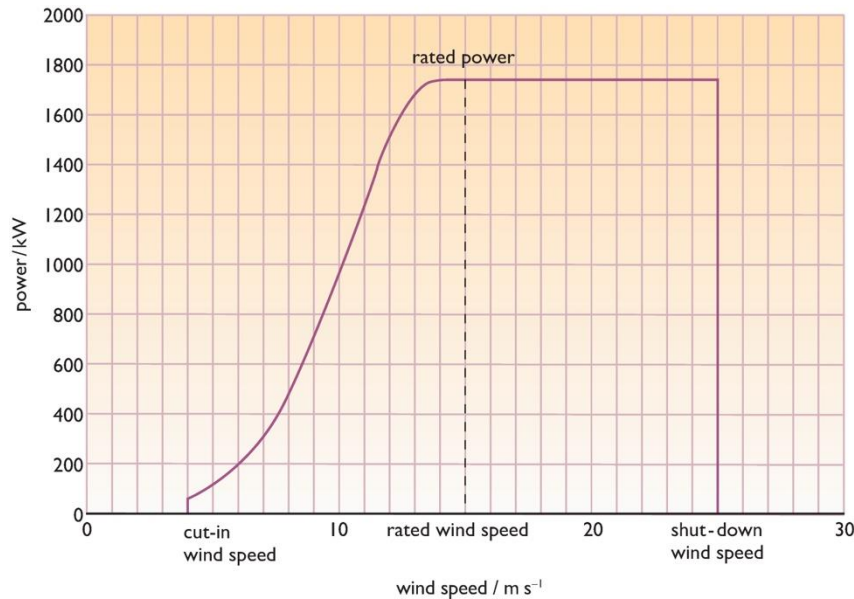
Idealized Wind Turbine Power Curve



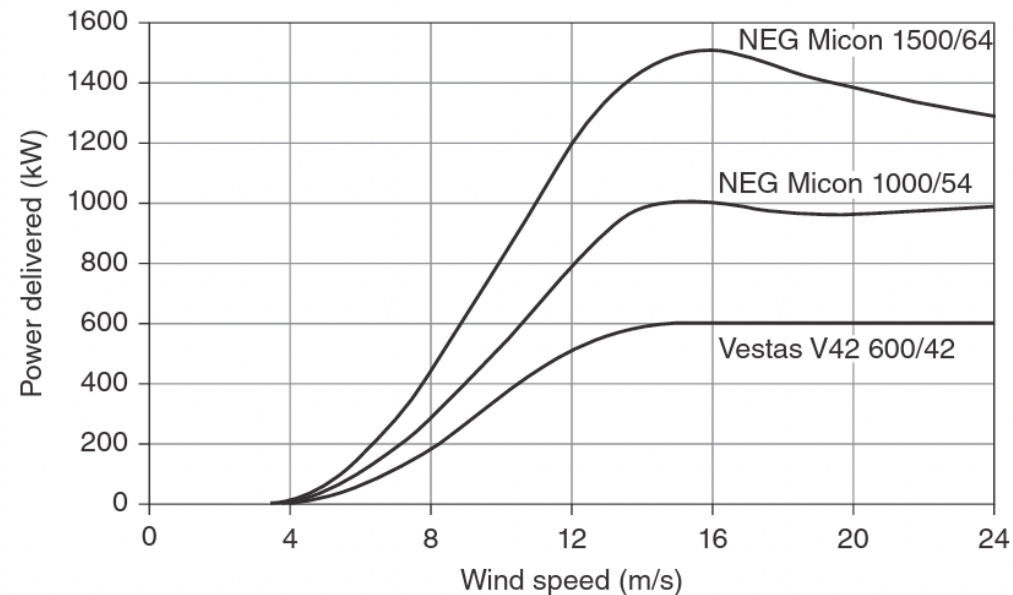
- V_C : the minimum needed to generate net power.
- V_R : the generator delivers as much as power as the it's designed for.
- V_F : machine is shut down for too strong wind (mechanical brakes lock the rotor shaft in place).

Real Examples

- Cut-in, rated, and shut-down wind speeds depend on turbine models. Typically V_C (4-8m/s), V_R (12-16m/s), V_F (24-28m/s).



Rated power 1500 kW &
rotor diameter 64 m



Matching Generators and Rotors

Example 7.4 Matching Generators and Rotors. An 82-m, 1.65-MW, fixed-speed wind turbine has a rated wind speed of 13 m/s. It is connected through a gearbox to a 4-pole, 60-Hz synchronous generator.

- What gear ratio should the gearbox have if the turbine is designed to turn at 14.4 rpm?
- What is the tip speed ratio when the winds are blowing at the rated wind speed?
- What is the overall efficiency of the machine (including rotor, gearbox, generator, etc.) at its rated wind speed?
- The power curve for this machine indicates that it will deliver half of its rated output in 8 m/s winds. What is its efficiency and TSR at that wind speed?
- What would be the TSR with 8 m/s winds if the generator could switch from four poles to six?

Solution

- From Equation 7.1 the generator shaft spins at

$$N = \frac{120f}{p} = \frac{120 \times 60}{4} = 1800 \text{ rpm}$$

So the gear ratio should be

$$\text{Gear ratio} = \frac{\text{Generator speed}}{\text{Rotor speed}} = \frac{1800 \text{ rpm}}{14.4 \text{ rpm}} = 125$$

Matching Generators and Rotors (cont'd)

b. The TSR at the 13 m/s rated wind speed:

$$\text{TSR} = \frac{82\pi \text{ m/rev} \times 14.4 \text{ rev/min}}{13 \text{ m/s} \times 60 \text{ s/min}} = 4.76$$

c. Overall efficiency at rated wind speed:

Assuming the standard 1.225 kg/m^3 air density, the power in the wind at 13 m/s is

$$P_w = \frac{1}{2} \rho A v_w^3 = \frac{1}{2} \times 1.225 \times \frac{\pi}{4} \times 82^2 \times 13^3 = 7106 \times 10^3 \text{ W}$$

So the overall efficiency of this 1.65 MW machine is

$$\text{Overall efficiency} = \frac{1650 \text{ kW}}{7106 \text{ kW}} = 23.2\%$$

Matching Generators and Rotors (cont'd)

d. Efficiency and TSR at 8 m/s:

$$P_w = \frac{1}{2} \times 1.225 \times \frac{\pi}{4} \times 82^2 \times 8^3 = 1656 \times 10^3 \text{ W}$$

$$\text{Overall efficiency} = \frac{0.5 \times 1650 \text{ kW}}{1656 \text{ kW}} = 49.8\%$$

$$\text{TSR} = \frac{82\pi \text{ m/rev} \times 14.4 \text{ rev/min}}{8 \text{ m/s} \times 60 \text{ s/min}} = 7.7$$

e. Switching from four- to six-pole operation:

$$N = \frac{120f}{p} = \frac{120 \times 60}{6} = 1200 \text{ rpm}$$

With our 125 gear ratio, the turbine would now spin at

$$\text{Rotor} = 1200 \text{ rpm} / 125 = 9.6 \text{ rpm}$$

And TSR would be

$$\text{TSR} = \frac{82\pi \text{ m/rev} \times 9.6 \text{ rev/min}}{8 \text{ m/s} \times 60 \text{ s/min}} = 5.1$$

This looks closer to the optimum range of TSRs.

Average Wind Speed

$$P_{\text{avg}} = \left(\frac{1}{2} \rho A v^3 \right)_{\text{avg}} = \frac{1}{2} \rho A (v^3)_{\text{avg}} \neq \frac{1}{2} \rho A (v_{\text{avg}})^3$$

$$v_{\text{avg}} = \frac{\sum_i [v_i \cdot (\text{hours at } v_i)]}{\sum \text{hours}} = \sum_i [v_i \cdot (\text{fraction of hours at } v_i)] = \sum_i [v_i \cdot \text{probability}(v = v_i)]$$

EX: During a 10h period, there're 3h of no wind, 3h at 5mph, and 4h at 10mph.

$$\begin{aligned} v_{\text{avg}} &= \frac{\text{Miles of wind}}{\text{Total hours}} = \frac{3 \text{ h} \cdot 0 \text{ mile/h} + 3 \text{ h} \cdot 5 \text{ mile/h} + 4 \text{ h} \cdot 10 \text{ mile/h}}{3 + 3 + 4 \text{ h}} \\ &= \left(\frac{3 \text{ h}}{10 \text{ h}} \right) \times 0 \text{ mph} + \left(\frac{3 \text{ h}}{10 \text{ h}} \right) \times 5 \text{ mph} + \left(\frac{4 \text{ h}}{10 \text{ h}} \right) \times 10 \text{ mph} = 5.5 \text{ mph} \end{aligned}$$

$$(v^3)_{\text{avg}} = \frac{\sum_i [v_i^3 \cdot (\text{hours at } v_i)]}{\sum \text{hours}} = \sum_i [v_i^3 \cdot (\text{fraction of hours at } v_i)] = \sum_i [v_i^3 \cdot \text{probability}(v = v_i)]$$

Wind Histogram

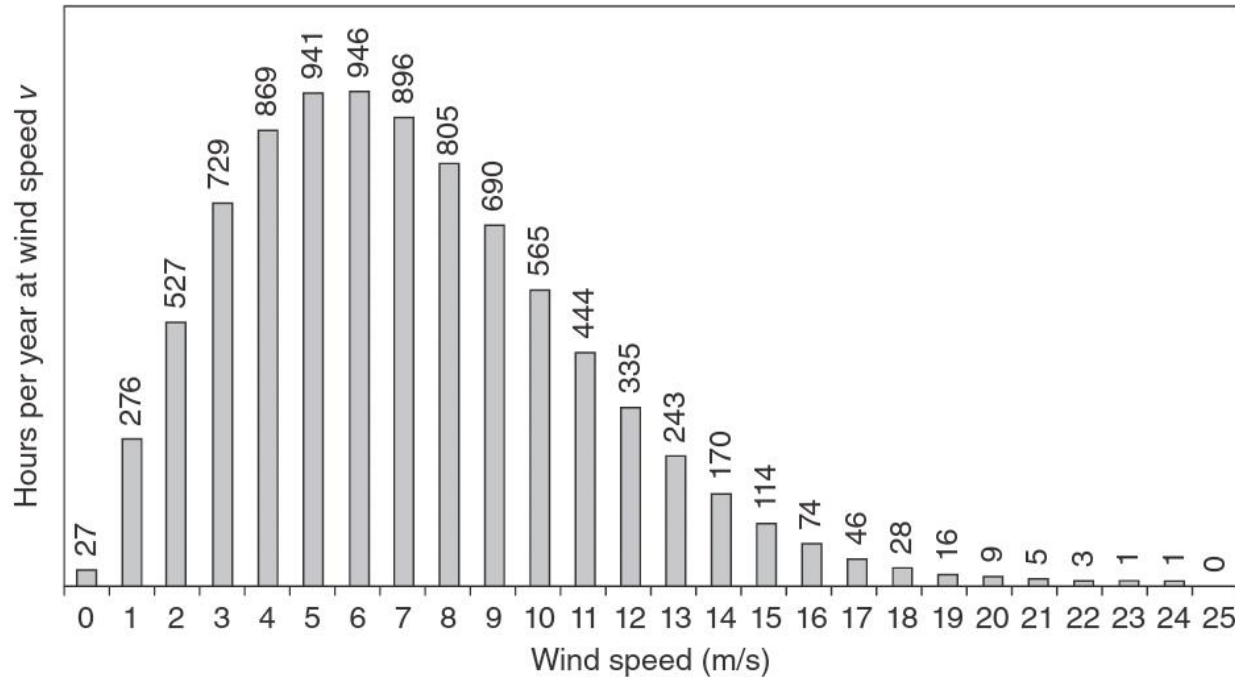


FIGURE 7.24 An example of a wind histogram showing hours that the wind blows in each wind speed bracket.

Example 7.5 Average Power in the Wind. Using the data given in Figure 7.24, find the average wind speed and the average power in the wind (W/m^2).

Assume the standard air density of 1.225 kg/m^3 . Compare the result with that which would be obtained if the average power were miscalculated using just the average wind speed.

Example Solution (1/2)

Solution. We need to set up a spreadsheet to determine average wind speed v and the average value of v^3 . Let us do a sample calculation of one line of a spreadsheet using 805 h/yr at 8 m/s:

$$\text{Fraction of the hours at 8 m/s} = \frac{805 \text{ h/yr}}{24 \text{ h/d} \times 365 \text{ d/yr}} = 0.0919$$

$$[v_8 \cdot (\text{Fraction of the hours at 8 m/s})] = 8 \times 0.0919 = 0.735$$

$$[(v_8)^3 \cdot (\text{Fraction of the hours at 8 m/s})] = 8^3 \times 0.0919 = 47.05$$

Wind Speed v_i (m/s)	Hours per Year at v_i	Fraction of Hours at v_i	$v_i \times$ Fraction of Hours at v_i	$(v_i)^3 \times$ Fraction of Hours at v_i
0	27	0.0031	0.000	0.00
1	276	0.0315	0.032	0.03
2	527	0.0602	0.120	0.48
3	729	0.0832	0.250	2.25
4	869	0.0992	0.397	6.35
5	941	0.1074	0.537	13.43
6	946	0.1080	0.648	23.33
7	896	0.1023	0.716	35.08
8	805	0.0919	0.735	47.05
9	690	0.0788	0.709	57.42
10	565	0.0645	0.645	64.50
.	.	.	.	.
.	.	.	.	.
22	3	0.0003	0.008	3.65
23	1	0.0001	0.003	1.39
24	1	0.0001	0.003	1.58
25	0	0.0000	0.000	0.00
Total:	8760	1.000	7.0	653.26

Example Solution (2/2)

The average wind speed is

$$v_{\text{avg}} = \sum_i [v_i (\text{Fraction of hours at } v_i)] = 7.0 \text{ m/s}$$

The average value of v^3 is

$$(v^3)_{\text{avg}} = \sum_i [(v_i)^3 \cdot (\text{Fraction of hours at } v_i)] = 653.24$$

The average power in the wind on a per unit of area basis is

$$P_{\text{avg}}/A = \frac{1}{2} \rho (v^3)_{\text{avg}} = 0.5 \times 1.225 \times 653.24 = 400 \text{ W/m}^2$$

If we had miscalculated average power in the wind by using the *average* wind speed of 7 m/s in Equation 7.7 we would have found:

$$P_{\text{avg}}/A(\text{WRONG}) = \frac{1}{2} \rho (v_{\text{avg}})^3 = 0.5 \times 1.225 \times 7.0^3 = 210 \text{ W/m}^2$$

From Histogram to PDF

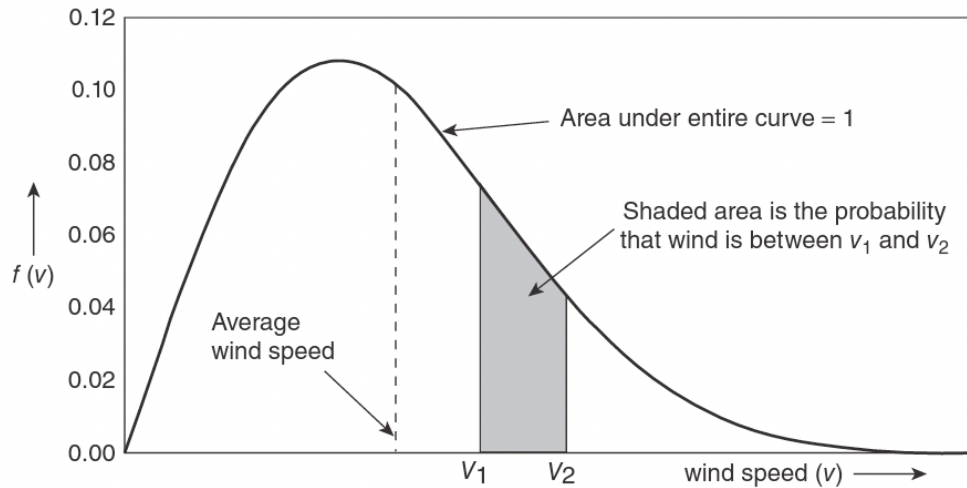


FIGURE 7.25 A wind speed probability density function (pdf).

$f(v)$ = wind speed probability density function

$$\text{Probability } (v_1 \leq v \leq v_2) = \int_{v_1}^{v_2} f(v)dv$$

$$\text{Probability } (0 \leq v \leq \infty) = \int_0^{\infty} f(v)dv = 1$$

The number of hours per year that wind blows between any two speeds:

$$\text{Hours/yr } (v_1 \leq v \leq v_2) = 8760 \int_{v_1}^{v_2} f(v)dv$$

$$v_{\text{avg}} = \int_0^{\infty} v \cdot f(v)dv$$

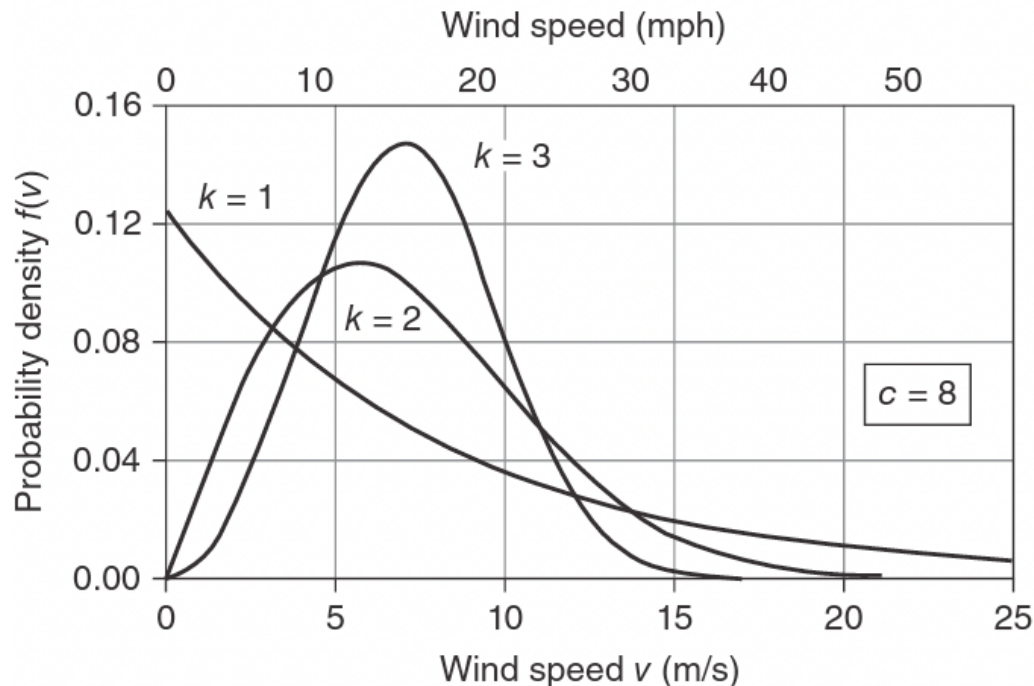
$$(v^3)_{\text{avg}} = \int_0^{\infty} v^3 f(v)dv$$

Wind Speed – Weibull Distribution

- Weibull distribution – probability density function (pdf)

$$f(v) = \frac{k}{c} \left(\frac{v}{c}\right)^{k-1} \exp\left[-\left(\frac{v}{c}\right)^k\right]$$

where $k > 0$ is the **shape** parameter, and $c > 0$ is the **scale** parameter.



Wind Speed – Rayleigh Distribution

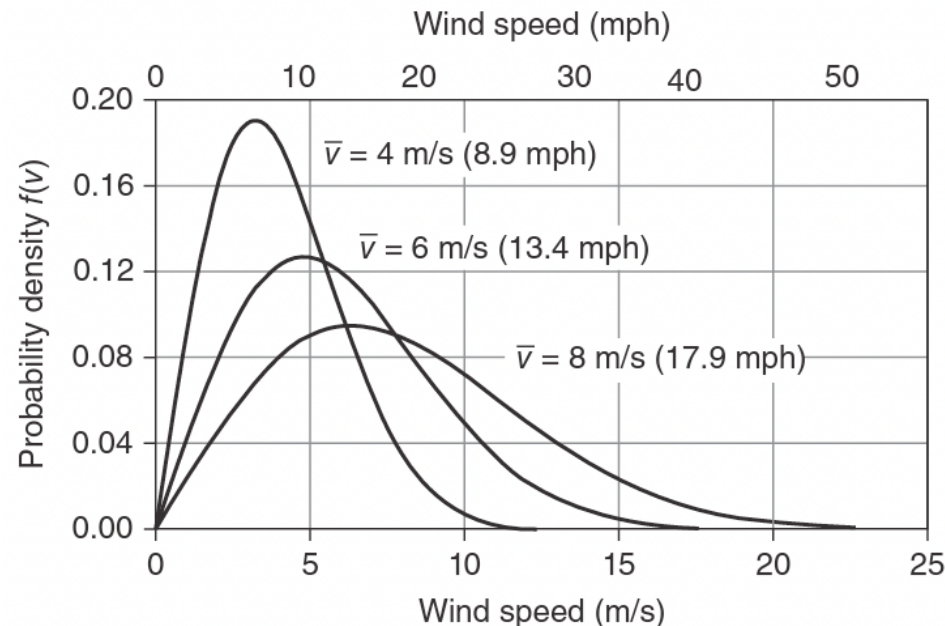
- Rayleigh distribution – a special case of Weibull: $k=2$

$$f(v) = \frac{2v}{c^2} \exp \left[- \left(\frac{v}{c} \right)^2 \right]$$

$$\bar{v} = \int_0^{\infty} v \cdot f(v) dv = \int_0^{\infty} 2 \left(\frac{v}{c} \right)^2 \exp \left[- \left(\frac{v}{c} \right)^2 \right] = \frac{\sqrt{\pi}}{2} c$$



$$f(v) = \frac{\pi v}{2\bar{v}^2} \exp \left[- \frac{\pi}{4} \left(\frac{v}{\bar{v}} \right)^2 \right]$$



Average Power for Wind with Rayleigh Distribution

$$\begin{aligned}(v^3)_{\text{avg}} &= \int_0^\infty v^3 \cdot f(v) dv = \int_0^\infty v^3 \cdot \frac{\pi v}{2\bar{v}^2} \exp\left[-\frac{\pi}{4} \left(\frac{v}{\bar{v}}\right)^2\right] dv \\ &= \frac{6}{\pi} \cdot \bar{v}^3 \approx 1.91\bar{v}^3\end{aligned}$$

$$\bar{P} = \frac{6}{\pi} \cdot \left(\frac{1}{2}\rho A \bar{v}^3\right) \text{ (Rayleigh assumptions)}$$

Example 7.6 Average Power in the Wind. A 10-m-high anemometer finds the average wind speed at that height to be 6 m/s. Estimate the average power in the wind at a height of 50 m. Assume Rayleigh statistics, a standard friction coefficient $\alpha = 1/7$, and standard air density $\rho = 1.225 \text{ kg/m}^3$. Repeat for a taller 80-m height.

Example Solution

Solution. We first adjust the winds at 10 m to those expected at 50 m using Equation 7.18:

$$\bar{v}_{50} = \bar{v}_{10} \left(\frac{H_{50}}{H_{10}} \right)^{\alpha} = 6 \left(\frac{50}{10} \right)^{1/7} = 7.55 \text{ m/s}$$

So, using Equation 7.49, per unit of area the average power in the wind would be

$$\bar{P} = \frac{6}{\pi} \times \left(\frac{1}{2} \rho A \bar{v}^3 \right) = \frac{6}{\pi} \left(\frac{1}{2} \times 1.225 \times 7.55^3 \right) = 504 \text{ W/m}^2$$

Let us take a slightly different approach to finding the power at 80 m. From Equation 7.20 we can write

$$\bar{P}_{10} = \frac{6}{\pi} \cdot \frac{1}{2} \cdot 1.225 \cdot 6^3 = 252.67 \text{ W/m}^2$$

$$\bar{P}_{80} = \bar{P}_{10} \left(\frac{H_{80}}{H_{10}} \right)^{3\alpha} = 252.67 \left(\frac{80}{10} \right)^{3 \times 1/7} = 616 \text{ W/m}^2$$

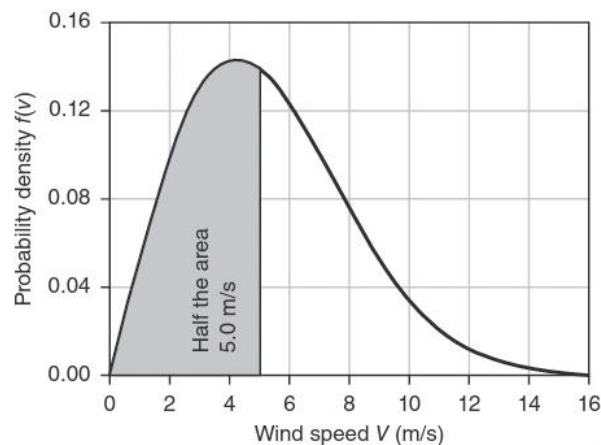
Wind Speed CDF

$$F(V) = \int_0^V f(v)dv = \text{prob}(v \leq V) = \int_0^V \frac{k}{c} \left(\frac{v}{c}\right)^{k-1} \exp\left[-\left(\frac{v}{c}\right)^k\right] dv$$

change of variable: $x = \left(\frac{v}{c}\right)^k \Rightarrow dx = \frac{k}{c} \left(\frac{v}{c}\right)^{k-1} dv$

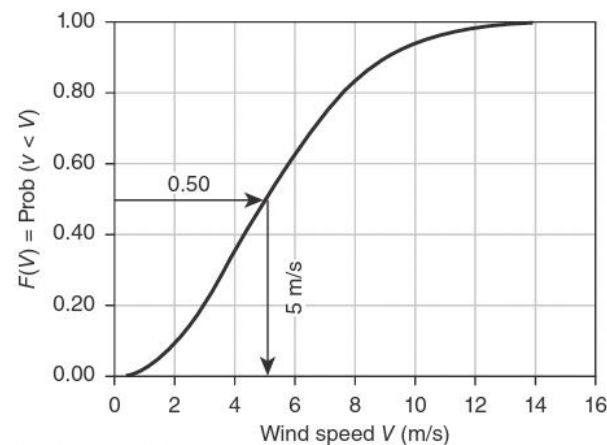
$$\Rightarrow F(V) = \int_0^{\left(\frac{V}{c}\right)^k} e^{-x} dx = 1 - \exp\left[-\left(\frac{V}{c}\right)^k\right] \quad (\text{Weibull})$$

Set $k = 2, c = 2\bar{v}/\sqrt{\pi} \Rightarrow F(V) = \text{prob}(v \leq V) = 1 - \exp\left[-\frac{\pi}{4} \left(\frac{V}{\bar{v}}\right)^2\right]$ (Rayleigh)



(a)

$k = 2$ and $c = 6$,



(b)

Example (1/2)

Example 7.7 Linking a Power Curve to Rayleigh Statistics. A 2.1-MW Suzlon S97 wind turbine has a cut-in wind speed $v_c = 3.5$ m/s, rated wind speed $v_R = 11$ m/s, and a furling wind speed of $v_F = 20$ m/s. If this machine is located in Rayleigh winds with an average wind speed of 7 m/s, find the following:

- How many hours per year is the wind below the cut-in wind speed?
- How many hours per year will the turbine be shut down due to excessive winds?
- How many kWh/yr will be generated when the machine is running at rated power?
- If this turbine in these winds has an overall capacity factor of 38%, what fraction of its annual energy delivered will be provided by winds below the 11 m/s rated wind speed?

Solution

- a. Using Equation 7.56, the probability that the wind speed is below cut-in 3.5 m/s is

$$\begin{aligned} F(V_c) &= \text{prob}(v \leq V_c) = 1 - \exp \left[-\frac{\pi}{4} \left(\frac{V_c}{\bar{v}} \right)^2 \right] \\ &= 1 - \exp \left[-\frac{\pi}{4} \left(\frac{3.5}{7} \right)^2 \right] = 0.178 \end{aligned}$$

In a year with 8760 h (365×24), the number of hours the wind will be less than 3.5 m/s is

$$\text{Hours}(v \leq 3.5 \text{ m/s}) = 8760 \text{ h/yr} \times 0.178 = 1562 \text{ h/yr}$$

- b. Using (7.58), the hours when the wind is higher than $V_F = 20$ m/s will be

$$\begin{aligned} \text{Hours}(v \geq V_F) &= 8760 \cdot \exp \left[-\frac{\pi}{4} \left(\frac{V_F}{\bar{v}} \right)^2 \right] \\ &= 8760 \cdot \exp \left[-\frac{\pi}{4} \left(\frac{20}{7} \right)^2 \right] = 14.4 \text{ h/yr} \end{aligned}$$

Example (2/2)

- c. Assuming its power curve is flat above V_R , the turbine will deliver 2100 kW any time the wind is between $V_R = 11$ m/s and $V_F = 20$ m/s. The number of hours that the wind is higher than 11 m/s is

$$\text{Hours}(v \geq 11) = 8760 \cdot \exp \left[-\frac{\pi}{4} \left(\frac{11}{7} \right)^2 \right] = 1260 \text{ h/yr}$$

So, the number of hours per year that the winds blow between 11 m/s and 20 m/s is $1260 - 14 = 1246$ h/yr. The energy the turbine delivers from those winds will be

$$\text{Energy}(V_R \leq v \leq V_F) = 2100 \text{ kW} \times 1246 \text{ h/yr} = 2.62 \times 10^6 \text{ kWh/yr}$$

With a 38% CF, the turbine will deliver

$$2100 \text{ kW} \times 8760 \text{ h/yr} \times 0.38 = 6.99 \times 10^6 \text{ kWh/yr}$$

Of that, we expect 2.62 million kWh/yr to be generated in winds above the rated wind speed, so the fraction delivered at winds below that will be

$$\% \text{ of energy from } v < 11 \text{ m/s} = \frac{6.99 - 2.62}{6.99} = 0.625 = 62.5\%$$